Lahore University of Management Sciences

AI 500: Foundations of AI

Fall 2026

Personnel Details	
Lead Instructor	**Hassan Mohy-ud-Din, PhD**
Room No. and Phone	9-246, SBASSE Tesla Wing. Telephone Extension: 8505
Office Hours	**Hassan:** appointment by email (hassan.mohyuddin@lums.edu.pk)
Co-Instructors	**TBA** **TBA** **TBA** **TBA** **TBA**
Co-Instructors' Office Hours	**TBA** **TBA** **TBA** **TBA** **TBA**
Course URL	LMS Site
Support Services	LUMS offers a range of academic and other services to support students. These are mentioned below, and you are encouraged to use these in addition to in-class assistance from course staff. For a complete list of campus support services available for you click here: https://advising.lums.edu.pk/#supportservices

Course Offering Details					
Credit Hours	3				
Classes (per week)	1	Duration	150 minutes	Timings & Venue	**Venue:** SDSB B-3; **Saturday:** 9:00 AM – 12.00 PM

Course Distribution	
Core	Yes
Elective	No
Open for Student Category	MS Artificial Intelligence (MS AI)

Course Description
Foundations of AI provides a comprehensive introduction to the fundamental concepts, methodologies, and practical workflows that underpin modern artificial intelligence and machine learning systems. The course develops a systems-level understanding of the complete AI pipeline, from problem formulation and data acquisition to model development, evaluation, deployment, and monitoring. Students build a strong foundation in the mathematical and computational principles required for AI, including linear algebra, probability, optimization, algorithmic complexity, and data representation. The course emphasizes data engineering practices such as data collection, cleaning, governance, feature engineering, and exploratory data analysis before introducing core machine learning paradigms, including supervised learning for regression and classification, unsupervised learning, dimensionality reduction, anomaly detection, and probabilistic reasoning under uncertainty. Students also learn principled approaches to model evaluation, interpretability, fairness, robustness, uncertainty quantification, and responsible model selection. Through applications drawn from healthcare, finance, engineering, business, public policy, and other domains, the course integrates theoretical foundations with practical AI system design, preparing students to develop reliable, interpretable, and ethically responsible AI solutions for real-world problems.

Assessed Course Learning Outcomes	
	The students should be able to:
CLO1:	Apply the fundamental concepts of AI, machine learning, probability, and optimization to formulate real-world AI problems.
CLO2:	Develop machine learning models using appropriate data preparation, feature engineering, and learning algorithms. Evaluate
CLO3:	AI models using appropriate performance metrics, validation techniques, and model interpretation methods.
CLO4:	Assess AI systems with respect to data quality, uncertainty, fairness, ethics, and responsible deployment.

Relation to EE Program Outcomes				
AI-620 CLOs	Related PLOs	Levels of Learning	Teaching Methods	CLO Attainment checked in
CLO1	-	-	Lectures, Assignments, Reading Material	Assignments/Quizzes/Exams
CLO2	-	-	Lectures, Assignments, Reading Material	Assignments/Quizzes/Exams
CLO3	-	-	Lectures, Assignments, Reading Material	Assignments/Quizzes/Exams
CLO4	-	-	Lectures, Assignments, Reading Material	Assignments/Quizzes/Exams

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Grading Breakup and Policy		
Assignments	20%	All assignments are compulsory.
Quizzes	25%	All quizzes are compulsory. Quizzes will be announced.
Midterm Exam	25%	Closed books, closed notes, In-person.
Final Exam	30%	Closed books, closed notes, In-person.

Assignment Grading Policy	
There will be **five to six programming assignments** throughout the semester.	
Grace Days and Late Submission	You are given a total of **5 grace days** to use across all assignments without penalty. Once these grace days are exhausted, late submissions will be penalized as follows: ● 1 day late : **20% penalty** (on that assignment) ● 2 days late : **40% penalty** (on that assignment) ● > 2 days late : **0% grade** (on that assignment)
Attendance Policy	Attendance records will be thoroughly maintained throughout the semester. A student is considered present if they attend at least 150 minutes of the 180 minutes of class. Each student is allowed **3 unexcused absences**. For each additional absence beyond the third, you **will lose one grace day** OR **lose 1% of points on a graded instrument** (excluding exam and project) if grace days are exhausted.

AI Policy

The use of generative AI tools (e.g., ChatGPT, Claude, etc.) in this course is strictly limited. You may use these tools only for the following purposes:

- **Searching and Information Gathering:** As a substitute for a search engine to look up definitions, formulas, syntax, notation, or explanations of specific concepts.
- **Clarification and Learning Support:** To obtain explanations of concepts, review documentation, or generate simple illustrative examples that help you understand course material.
- **Error Identification:** To identify and correct minor syntax, notation, formatting, or computational errors in work that you have independently produced.

All other uses of generative AI are prohibited. In particular, you may not:

- Generate or substantially develop the core logic, algorithms, mathematical derivations, proofs, solutions, or overall structure of assignments.
- Submit any code, mathematical work, written solutions, proofs, or other assignment content that is wholly or partially generated by AI.
- Use AI to complete graded work in place of your own reasoning, problem-solving, or implementation.

All submitted work must represent your own independent understanding and effort. Any unauthorized use of generative AI constitutes academic misconduct and will be handled in accordance with the institution's academic integrity policies. It is your responsibility to understand and comply with this policy.

Examination Detail	
Exam	**Comprehensive** **Mode:** Closed book, closed notes, In-person. A formula sheet will be provided with the question paper. **Duration:** 2 – 3 hours **Time and Date:** Announced in advance.

Campus Support and Key University Policies

Campus Supports

Students are strongly encouraged to meet course instructors and TA's during office hours for assistance in course-content, understand the course's expectations from enrolled students, etc. Beyond the course, students are also encouraged to use a variety of other resources. (Instructors are also encouraged to refer students to these resources when needed.) These resources include Counseling and Psychological Services/CAPS (for mental health), LUMS Medical Center/LMC (for physical health), Office of Accessibility & Inclusion/ OAI (for long-term disabilities), advising staff dedicated to supporting and guiding students in each school, online resources, etc. To view all support services, their specific role as well as contact information click here.

Academic Honesty/Plagiarism

LUMS has zero tolerance for academic dishonesty. Students are responsible for upholding academic integrity. If unsure, refer to the student handbook and consult with instructors/teaching assistants. To check for plagiarism before essay submission, use similarity@lums.edu.pk. Consult the following resources: 1) Academic and Intellectual Integrity and 2) Understanding and Avoiding Plagiarism.

LUMS Academic Accommodations/Petitions Policy

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Long-term medical conditions are accommodated through the Office of Accessibility & Inclusion (OAI). Short-term emergencies that impact studies are either handled by the course instructor or Student Support Services (SSS). For more information, please see Missed Instrument or "Petition" FAQs for students and faculty at https://rb.gy/8sj1h.

LUMS Sexual Harassment Policy

LUMS and this class are a harassment-free zone. No behavior that makes someone uncomfortable or negatively impacts the class or individual's potential will be tolerated. To report sexual harassment experienced or observed in class, please contact me. For further support or to file a complaint, contact OAI at oai@lums.edu.pk or harassment@lums.edu.pk. You may choose to file an informal or formal complaint to put an end to the offending behavior. You can also call their Anti-Harassment helpline at 042-35608877 for advice or concerns. For more information: *Harassment, Bullying & Other Interpersonal Misconduct: Presentation*.

SSE Council on Equity and Belonging

In addition to LUMS resources, SSE's Council on Belonging and Equity is committed to devising ways to provide a safe, inclusive and respectful learning environment for students, faculty and staff. To seek counsel related to any issues, please feel free to approach either a member of the council or email at cbe.sse@lums.edu.pk.

Rights and Code of Conduct for Online Teaching

The misuse of online modes of communication is unacceptable. TAs and Faculty will seek consent before the recording of live online lectures or tutorials. Please ensure, if you do not wish to be recorded during a session, to inform the faculty member. Please also ensure that you prioritize formal means of communication (email, LMS) over informal means to communicate with course staff. Kindly follow the copyright rules attached with the educational materials uploaded on LMS by the instructor.

Detailed Lecture Plan

Module	Topics	Content
1	**The AI/Data Science Landscape and Problem Framing**	• What is AI? Difference between AI, machine learning, data science, statistics, analytics, optimization, automation, and generative AI. • AI as a pipeline: **problem → data → representation → model → evaluation → decision → deployment → monitoring** • Types of AI problems: prediction, classification, regression, ranking, recommendation, clustering, search, planning, control, anomaly detection, generation, causal estimation. • Translating real-world problems into AI tasks: inputs, outputs, labels, constraints, risks, stakeholders, cost of error. • Examples from health, finance, business, public policy, climate, education, and engineering. • Why many AI projects fail: wrong problem framing, poor data, leakage, weak evaluation, lack of deployment thinking.
2	**Mathematical and Computational Foundations for AI**	• Data representation: scalars, vectors, matrices, tensors, tables, graphs, text, images, time series. • Linear algebra essentials: vector spaces, dot products, norms, projections, matrix multiplication, rank, eigenvectors, SVD intuition. • Probability essentials: random variables, distributions, expectation, variance, conditional probability, Bayes rule. • Optimization basics: objective functions, gradients, convex vs non-convex problems, empirical risk minimization. • Data structures for AI: arrays, hash maps, trees, heaps, graphs, sparse matrices. • Algorithmic complexity: Big-O, memory vs compute trade-offs, why scalability matters in AI.
3	**Data Collection, Curation, Cleaning, Governance, and Ethics**	• Data sources: surveys, sensors, web, logs, administrative records, experiments, public datasets, synthetic data. • Data quality dimensions: completeness, accuracy, consistency, timeliness, representativeness, measurement error. • Data cleaning: missing values, outliers, duplicates, inconsistent labels, unit errors, class imbalance. • Data leakage: target leakage, train-test contamination, temporal leakage, group leakage. • Dataset documentation: datasheets, data dictionaries, provenance, consent, licensing. • Governance and ethics: privacy, bias, fairness, consent, data minimization, sensitive attributes, institutional approval. • Practical case: how a seemingly "good" dataset can produce a misleading AI system.
4	**Exploratory Data Analysis, Feature Engineering, and Statistical Thinking**	• EDA goals: understanding distributions, relationships, anomalies, missingness, bias, and subgroup behavior. • Visualization: histograms, boxplots, scatterplots, correlation heatmaps, pair plots, time-series plots, categorical summaries. • Descriptive statistics: mean, median, variance, quantiles, skewness, covariance, correlation. • Feature engineering: scaling, normalization, binning, encoding categorical variables, interaction features, polynomial features, time-window features. • Statistical foundations: samples vs populations, sampling bias, variance, uncertainty, confidence intervals. • Hypothesis testing: null/alternative hypotheses, p-values, effect size, statistical vs practical significance. • Common mistakes: p-hacking, multiple comparisons, correlation mistaken for causation.

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5	**Supervised Learning I: Regression, Loss Functions, and Generalization**	• Supervised learning setup: dataset, features, labels, model, parameters, prediction function, loss function. • Regression problems: linear regression, polynomial regression, regularized regression. • Loss functions: MSE, MAE, Huber loss; when each is appropriate. • Training vs testing: empirical risk, population risk, overfitting, underfitting. • Bias-variance trade-off and model capacity. • Regularization: L_1, L_2, early stopping intuition. • Cross-validation and hyperparameter tuning.
6	**Supervised Learning II: Classification and Core ML Algorithms**	• Classification setup: binary, multiclass, multilabel classification. - Logistic regression as probabilistic classification. • k-nearest neighbors, decision trees, random forests, gradient boosting. • Probabilistic classifiers: Naive Bayes and likelihood-based reasoning. • Class imbalance: resampling, class weights, threshold adjustment. • Multiclass strategies: one-vs-rest, one-vs-one, softmax-style classification. • Practical comparison: when simple models outperform complex models.
7	**Model Evaluation, Error Analysis, Interpretability, and Responsible Model Selection**	• Evaluation metrics for regression: RMSE, MAE, R^2, calibration-style error interpretation. • Evaluation metrics for classification: accuracy, precision, recall, F_1, ROC-AUC, PR-AUC, confusion matrix. • Threshold selection: cost-sensitive decisions and domain-specific trade-offs. • Calibration: probability outputs vs hard labels. • Error analysis: subgroup errors, failure modes, data slices, false positives vs false negatives. • Interpretability: coefficients, feature importance, permutation importance, partial dependence, SHAP/LIME intuition. • Robustness and fairness basics: distribution shift, subgroup performance, fairness metrics. • Choosing a model responsibly: performance, simplicity, cost, explainability, risk.
8	**Unsupervised Learning, Dimensionality Reduction, and Anomaly Detection**	• Why unsupervised learning matter when labels are unavailable or expensive. • Clustering: k-means, hierarchical clustering, DBSCAN intuition. • Cluster evaluation: silhouette score, stability, interpretability, domain validation. • Dimensionality reduction: PCA, SVD, t-SNE/UMAP intuition and limitations. • Representation learning intuition: from raw features to compact latent structure. • Anomaly detection: statistical thresholds, distance-based methods, isolation forests, reconstruction-error intuition. • Applications: fraud detection, sensor faults, customer segmentation, image/text exploration.
9	**Probabilistic AI: Uncertainty and Bayesian Reasoning**	• Why AI must reason under uncertainty: noisy data, incomplete information, ambiguous evidence. • Bayesian perspective: priors, likelihood, posterior, evidence. • Naive Bayes revisited as a simple probabilistic model. • Why uncertainty quantification matters; aleatoric, epistemic, model, data, measurement, and distributional uncertainty. • Decision-making under uncertainty: expected utility and risk. • Predictive distributions and uncertainty summaries; bootstrapping, Bayesian inference, Monte Carlo dropout, and confidence intervals. • Evaluation and deployment: calibration, expected calibration error (ECE), Brier score, negative log-likelihood (NLL), and out-of-distribution detection. • Applications: diagnosis, forecasting, user behavior, sensor fusion.
10	**Graphical Models and Causal Inference** *(if time permits)*	• An introduction to Graphical Models with applications. • An Introduction to Causal Machine Learning with applications.

REMARKS: The 10-module lecture plan will be covered throughout the semester. As indicated in the plan, the course emphasizes Data Engineering, Pipelines, and ML Systems.

Recommended Reading Material

Books on Machine Learning

- Laurent Younes, **Introduction to Machine Learning**, cis.jhu.edu, (2025). [Ebook]
- Sebastien Roch, **Mathematical Methods in Data Science**, Cambridge University Press, (2025). [Ebook]
- Ruye Wang, **Introduction to Machine Learning, From Math to Code**, Cambridge University Press, (2025). [Ebook]
- Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, and Jonathan Taylor, **An Introduction to Statistical Learning with Applications in Python**, Springer Cham, (2023). [Ebook]
- Kevin P. Murphy, **Probabilistic Machine Learning: Advanced Topics**, The MIT Press, (2022). [Ebook]
- Kevin P. Murphy, **Probabilistic Machine Learning: An Introduction**, The MIT Press, (2022). [Ebook]
- Ali H. Sayed, **Inference and Learning from Data, Volumes I, II, III**, Cambridge University Press, (2022).
- Stuart Russell and Peter Norvig, **Artificial Intelligence: A Modern Approach**, Pearson Publishers, (2022). [Ebook]
- Andrew Wolf, **Machine Learning Simplified: A Gentle Introduction to Supervised Learning**, mlsysbook.ai, (2022). [Ebook]
- Jeremy Watt, Reza Borhani, and Aggelos K. Katsaggelos, **Machine Learning Refined**, Cambridge University Press, (2020). [Ebook]

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- Kevin P. Murphy, **Machine Learning, A Probabilistic Perspective**, The MIT Press, (2012). [Ebook]
- Ethem Alpaydın, **Introduction to Machine Learning**, The MIT Press, (2010). [Ebook]
- Alex Smola and S.V.N. Vishwanathan, **Introduction to Machine Learning**, Cambridge University Press, (2008).
- Christopher M. Bishop, **Pattern Analysis and Machine Learning**, Springer, (2006). [Ebook]

Books on Deep Learning

- Christopher M. Bishop and Hugh Bishop, **Deep Learning: Foundations and Concepts**, Springer (2024). [Ebook]
- Simon J. D. Prince, **Understanding Deep Learning**, MIT Press, (2024). [Ebook]
- Francois Fleuret, **The Little Book of Deep Learning**, Creative Commons, (2023). [Ebook]
- Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, **Dive into Deep Learning**, arXiv, (2021). [Ebook]
- Francois Chollet, **Deep Learning with Python**, Manning Publications, (2018).
- Michael Nielsen, **Neural Networks and Deep Learning**, Determination press, (2015). [Ebook]

Books on Optimization

- Ali H. Sayed, **Inference and Learning from Data, Volumes I, II, III**, Cambridge University Press, (2022).
- Jorge Nocedal and Stephen Wright, **Numerical Optimization**, Springer Science & Business Media, (2006).
- Stephen P. Boyd and Lieven Vandenberghe, **Convex Optimization**, Cambridge University Press, (2004). [Ebook]
- Dimitri P. Bertsekas, **Nonlinear Programming**, Athena Scientific, (1999).

Books on Python Programming

- Kyle Gallatin and Chris Albon, **Machine learning with Python Cookbook**, O'Reilly Media, Inc, (2023).
- Bernd Klein, **Machine Learning with Python Tutorial**, Alanna Maldonado, (2023).
- Joakim Sundnes, **Introduction to Scientific Programming with Python**, Springer Nature, (2020).
- Luciano Ramalho, **Fluent Python: Clear, Concise, and Effective Programming**, O'Reilly Media, Inc, (2023).
- Teik T. Teoh and Zheng Rong, **Artificial Intelligence with Python**, Springer, (2022).
- Jake VanderPlas, **Python Data Science Handbook: Essential tools for working with data**, O'Reilly Media, Inc, (2016).
- Andreas C. Müller and Sarah Guido, **Introduction to Machine Learning with Python: A guide for Data Scientists**, O'Reilly Media, Inc, (2016).
- Bill Lubanovic, **Introducing Python: Modern Computing in Simple Packages**, O'Reilly Media, Inc, (2019).
- Jacqueline Kazil and Katharine Jarmul, **Data Wrangling with Python: Tips and tools to make your life easier**, O'Reilly Media, Inc, (2016).

Prepared and revised by:	**Dr. Hassan Mohy-ud-Din**
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